Ricardo Neves

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Summary

Staff Software Engineer with 10+ years of experience building scalable, resilient, and high-performance systems at growing companies and global orgs. At Adyen, I have worked across Payments, Platform Engineering, and Unified Platform, where I now scale the core ledger for transactional and accounting data. Previously, I was the first hire in Klue's Amsterdam office and an early engineer at Defined.ai, helping it grow from 7 to 150 employees in three years.

Experience

Staff Software Engineer Sep 2021 – Present

Adyen

- Led the core ledger's scaling to support **20–30% year-over-year growth** and up to **199k payment transactions per minute** while maintaining **99.99% availability**
- Led the ledger platform's re-architecture to support expansion into new availability zones
- Led a DevEx program to speed up delivery and reduce the core accounting group's change failure rate
- Helped establish AI engineering practices through org-wide knowledge sharing and contributions to tooling
- Led database capacity planning through stress testing and performance analysis, and designed application, database, and network experiments for new data-center rollouts
- Established a new 24/7 on-call rotation for Unified Platform and mentored its engineers
- Built tooling to simplify the migration of legacy queue-like workloads, later adopted by other teams
- Led work on the internal database framework to enable real-time connection tuning and improve observability
- Improved the resilience of the distributed locking framework by adding liveness checks
- Shared expertise across the org through talks and writing on topics ranging from SLOs to Postgres query tuning

Tech: Java, Postgres, MyBatis, HikariCP, Prometheus, ELK, Nomad

Software Engineer Apr 2020 – Sep 2021

Adyen

- Led the development of a multi-region, real-time platform for in-person payments and fleet management, scaling it from thousands to **100k+ devices** while maintaining low latency and high availability
- Designed a stress-test suite to identify scaling bottlenecks and support capacity planning for projected growth
- Improved monitoring and alerting using black-box checks and internal telemetry
- Led the creation of a new environment to accelerate firmware integration and compliance certification

Tech: Java, Postgres, MyBatis, Tomcat, ELK

Software Engineer Aug 2019 – Apr 2020

Klue

- Helped establish Klue's first overseas office with an early employee and participated in hiring from day one
- Set up Google Cloud infrastructure for Python services, including CI/CD, load balancers, TLS, and Kubernetes
- Upgraded Elasticsearch by refactoring the application and migrating production data
- Improved application security and resiliency with rate limiting, user permissions, and subresource integrity
- Built a nearest-neighbor deduplication pipeline processing **1M+ articles and reviews per day**

Tech: Python, Ruby, Flask, Rails, Elasticsearch, Redis, Docker, Kubernetes

Lead Software Engineer Jan 2019 – Jul 2019

Defined.ai

- Led engineering for Data Infrastructure and Machine Learning
- Managed the quarterly roadmap, headcount planning, and hiring
- Architected the first real-time and batch quality-assessment pipelines handling **1M+ crowdsourcing tasks per day**
- Built frameworks to accelerate machine learning development and deployment
- Set up infrastructure and provided tools for exploratory research, business analysis, and reporting

Tech: Python, C#, Azure Service Bus, Redis, Cosmos DB, Hive, Spark, Azure Data Lake

Software Engineer Sep 2016 – Dec 2018

Defined.ai

- Built an enterprise SaaS catalog and pricing backend
- Built a text translation backend and reusable frontend components
- Added support for new crowdsourcing tasks used in 56+ countries, including localization
- Shipped iOS and Android audio collection apps with offline support
- Set up CI/CD and infrastructure for machine learning APIs

Tech: C#, ASP.NET, SQL Server, React, Redux, Swift, Java, React Native, TensorFlow, Docker, Kubernetes

Software Consultant Oct 2015 – Aug 2016

NTT Data (previously Everis)

- Improved the daily financial reporting pipeline for a capital markets platform from hours to seconds
- Expanded integration between client systems and a centralized identity and access management solution using Active Directory

Tech: Java, MySQL, JBoss, Active Directory, CA Identity Suite

FCT Research Scholarship Jan 2015 – Sep 2015

INESC-ID

- Researched energy-efficient transactional memory using machine-learning-guided self-tuning for the Green-TM project

Tech: C, C++, Java, Weka

Projects

Lintmatter | *Rust* 2026

- A linter for agentic context

Tea | *Python, Docker* 2026

- Testing and evaluation framework for agentic workflows

Silo | *Rust, Ratatui* 2026

- Isolated workspace manager for parallel agentic development

Chillout | *Vue.js, Node.js, PeerJS* 2020

- Peer-to-peer video chat app to hangout with friends

Koopera | *Python, Flask, Vue.js* 2019

- Collaboration app for sharing and reviewing Jupyter notebooks

Education

Instituto Superior Técnico, Lisbon, Portugal 2013 – 2015

M.Sc. in Software Engineering and Distributed Systems

Publications

ProteusTM: Abstraction Meets Performance in Transactional Memory | *ASPLOS 2016*

- Didona, Diegues, Kermarrec, Guerraoui, Neves, Romano